INSTITUTE OF TECHNOLOGY BLANCHARDSTOWN

Academic term	2018/2019
Year of study	2
Semester	SEMESTER ONE
Date of examination	Fri 4th Jan 2019
Time of examination	3.30am – 5.30pm

Programme code	Programme title	Module code
BN002	Higher Certificate in Science in Computing in Information Technology	COMP H2026
BN013	Bachelor of Science in Computing in Information Technology	COMP H2026
BN034	Bachelor of Science in Computing in Digital Forensics & Cyber Security	COMP H2026
BN104	Bachelor of Science (Honours) in Computing	COMP H2026
BN120	Bachelor of Science (Honours) in Computing in Digital Forensics & Cyber Security	COMP H2026

Module title	Information Technology Mathematics

Internal Examiner(s)	**Laura Keyes**
External Examiner(s)	**Dr. Christopher Meudec** **Dr. Simon Caton**

Instructions to candidates:

1.	**To ensure that you take the correct examination, please check that the module and programme which you are following is listed in the table above.**
2.	**Question 1 is compulsory. Candidates should attempt ALL parts of Question 1 and any other TWO questions.**
3.	**This paper is worth 100 marks. Question 1 is worth 40 marks. The remaining questions are worth 30 marks each.**
4.	**Remember to mark each number and section clearly and to list the questions which you have attempted in the correct order on the front of your exam answer script.**
5.	**Show all your work.**

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Question 1: (Compulsory)

Attempt ALL Parts

a) Given the following matrices:

$$A = \begin{bmatrix} 2 & -3 \\ 0 & 2 \\ 7 & -2 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & -2 & 0 \\ 5 & 1 & 2 \end{bmatrix}$$

i. What is the size of each matrix *A* and *B*?
ii. Write down the value of the elements a_{32} and b_{12}.
iii. Write down the transpose of matrix *B*.
iv. Find the matrix given by $2(A + B^T)$.

(5 marks)

b) Determine the *inverse* of $\begin{bmatrix} 5 & -2 \\ -7 & 3 \end{bmatrix}$. **(5 marks)**

c) Answer the following questions based on the rooted tree shown below in figure 1.1.

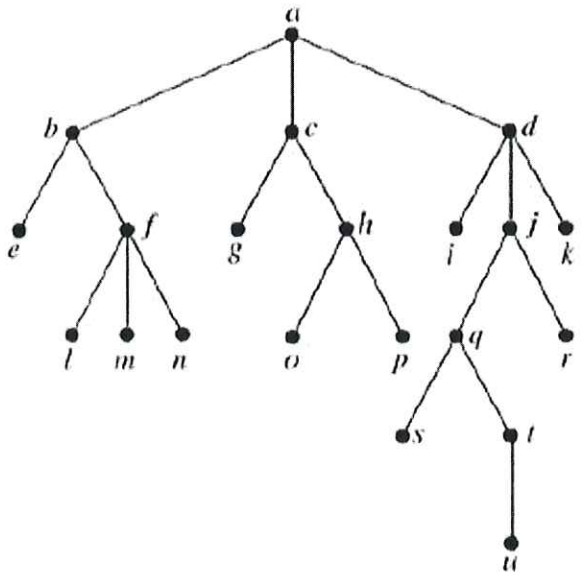


Figure 1.1 Rooted Tree

i. Which node is the *root*?
ii. Which nodes are children of *c*.
iii. Which node is the parent of *j*.
iv. Find the number of *leaf* nodes in this rooted tree.
v. List the ancestors of node *q*.

(5 marks)

d) Draw the graph *G* represented by the following adjacency matrix and write down the degree of each vertex in *G*.

$$\begin{pmatrix} 2 & 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

(5 marks)

e) Using the dataset 3, 7, 15, 10, 8, 17, 1, 20
   i. Mean
   ii. Standard Deviation

(5 marks)

f) If two fair dice are thrown:
   i. What is the probability of getting a three and a four?
   ii. What is the probability that the SUM of the numbers will be greater than or equal to four.

(5 marks)

g) Using differentiation from **first principles**, find the derivative of $y = 4x^2 - 2x$.

(5 marks)

h) Evaluate the following *indefinite* integrals:
   i. $\int (12x^2 + 6x + 3).dx$
   ii. $\int (x^{-2}).dx$

(5 marks)

Total: 40 marks

Question 2: Differentiation and Integration

a) Figure 2.1 below shows the graph of $y = x^3 - 12x + 25$. *A* is the maximum point of the curve. *C* is the minimum point of the curve. The curve crosses the *y*-axis at *B*, that is, where $x = 0$.

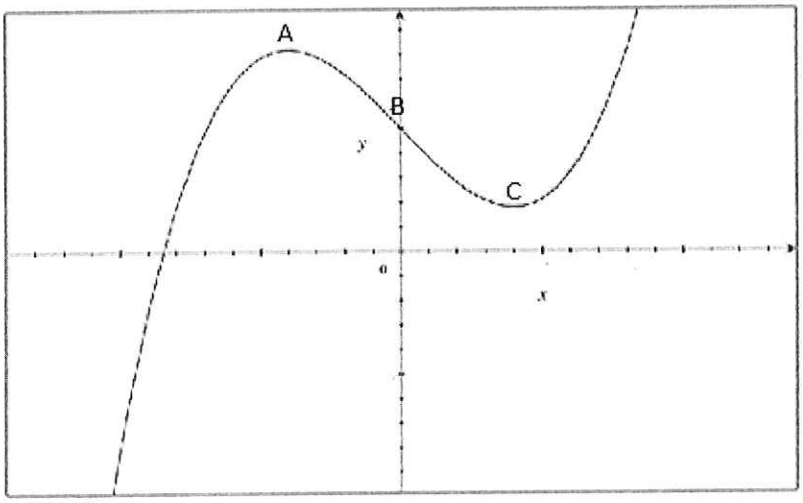


Figure 2.1 Curve given by $y = x^3 - 12x + 25$

i. Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$. **(4 marks)**

ii. Find the slope of the tangent to the curve at the point *B*. **(2 marks)**

iii. Find the coordinates of the maximum and minimum turning points of the curve, *A* and *C* respectively. **(5 marks)**

b) Differentiate the following:

i. $y = \frac{3x}{2x+1}$ **(4 marks)**

ii. $y = (4x^2 + 9)^3$ **(4 marks)**

c) A circuit is designed so that it outputs an alternating current (*I*) whose value in milliamps at any time *t* seconds after the circuit is activated is given by: $1 + 30t - 5t^2$. Find,

i. The current 1 second after the circuit is activated.

ii. The rate at which the current is changing after 2 seconds.

iii. The time taken for the current to reach its first peak value. **(5 marks)**

d) Find the area enclosed by the curve $y = x + 2x^2$ for x between 0 and 2. **(6 marks)**

Total: 30 marks

Question 3: Statistics and Probability

a) Use the scatter plot in figure 3 below to answer the following questions.

i. What is meant by *correlation* in statistics? What type of correlation is shown in figure 3.1? **(3 marks)**

ii. How would you identify any *exceptions* in the data? In your answer discuss any *exceptions* in the data represented in figure 3.1.

(3 marks)

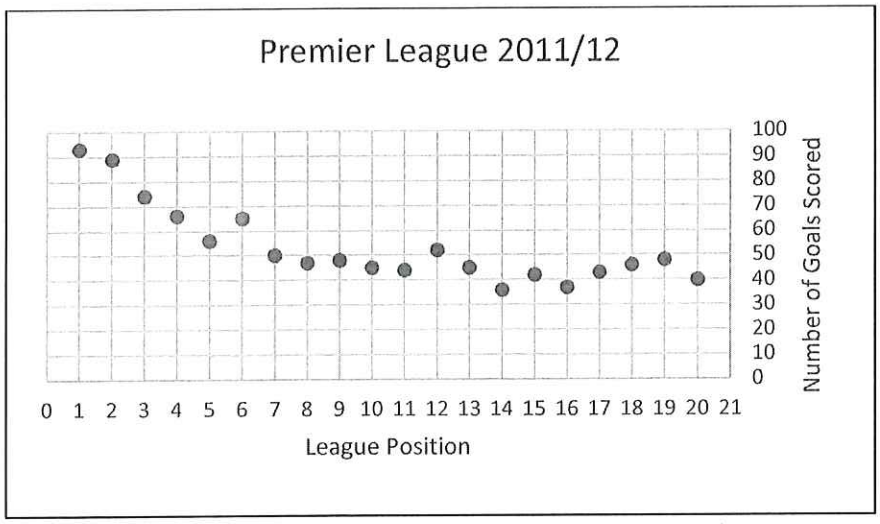


Figure 3.1 Scatter plot of League position and goals scored

Question continued over page.

b) The following table (Table 3.1) is the distribution of the number of months a group of 27 students have owned their current mobile phone.

Table 3.1

Number of Months	Frequency (No. of Students)
$0 \leq x < 4$	4
$4 \leq x < 8$	2
$8 \leq x < 12$	3
$12 \leq x < 16$	3
$16 \leq x < 20$	4
$20 \leq x < 24$	4
$24 \leq x < 28$	3
$28 \leq x < 32$	1
$32 \leq x < 36$	2
$36 \leq x < 40$	1

i. Calculate the *Mean* of the grouped data. **(3 marks)**
ii. Calculate the *Standard Deviation* of the grouped data. **(5 marks)**
iii. Draw a h*istogram* for the data. **(4 marks)**
iv. Comment on the *symmetry* or *skewness* of the distribution. **(2 marks)**

c) A local sports club holds a weekly lottery draw based around picking 4 correct numbers from 32. What is the probability of winning in this lotto draw?

(3 marks)

d) Mary uses her mobile phone for X minutes each day. *X* is a random variable which may be modelled by a normal distribution, with a mean of 28 minutes and a standard deviation of 8 minutes. What is the probability that on a particular day, Mary uses her phone for between 10 and 40 minutes?

(7 marks)

Total: 30 Marks

Question 4: Matrices, Graphs and Trees

a) Given the following matrices:

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 4 & 5 & 3 \\ 2 & 2 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 3 & -1 & 4 \\ -2 & 3 & 1 \\ 2 & 4 & -1 \end{bmatrix} \quad C = \begin{bmatrix} 2 \\ 4 \\ 3 \end{bmatrix}$$

i. Write down the Identity matrix for B **(3 marks)**

ii. Calculate $A^T C$ **(3 marks)**

iii. Calculate the *inverse* of matrix *B* **(9 marks)**

b) For the graph in figure 4.1 below, find a Hamiltonian circuit or explain why no Hamiltonian circuit exits.

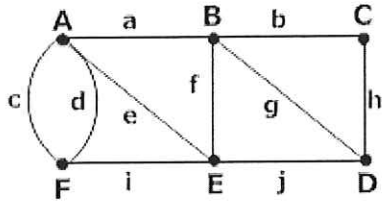


Figure 4.1

(3 marks)

c) Describe the difference between *breadth-first* and *depth-first* graph traversals. Consider the following graph (figure 4.2), starting at node *A*, in what order will the nodes be visited using *Breadth First Search*? Assume, that if there is a choice between multiple nodes, the algorithm will choose alphabetically. Show all your work.

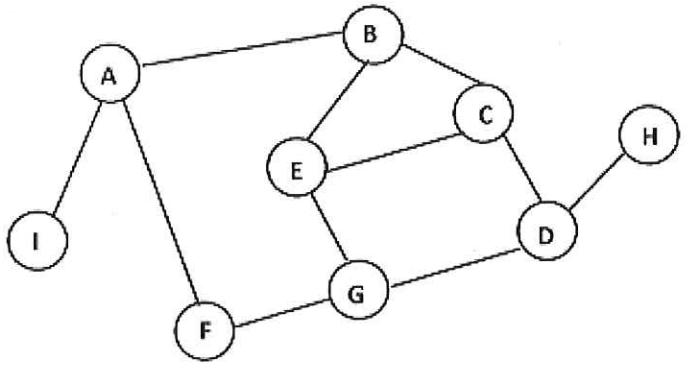


Figure 4.2

(12 marks)

Total: 30 Marks

Formulae

Matrix Determinant

$\det A = ad - bc$ (2x2 matrix)

$\det A = a_{11}c_{11} + a_{12}c_{12} + a_{13}c_{13}$ (3x3 matrix using row 1)

Matrix Inverse

$$A^{-1} = \frac{1}{\det A}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$ (2x2 matrix)

$$A^{-1} = \frac{1}{\det A}C^T$$ (3x3 matrix)

General Probability:

If two events A and B are mutually exclusive, then the following law holds:

$P[A \text{ or } B] = P[A] + P[B]$

If the two events are not mutually exclusive, the law above is broadened so that the probability of A or B occurring is given by:

$P[A \text{ or } B] = P[A] + P[B] - P[A \text{ and } B]$

Consider two events *A* and *B* with event *B* dependent on *A*. The probability of the occurrence of both events is given by:

$P[A \text{ and } B] = P[A] \times P[B \mid A]$

Probability Distributions:

$P[X = r] = {}^nC_r p^r q^{n-r} = \frac{n!}{(n-r)!r!} p^r q^{n-r}$ (Binomial distribution)

$P[X = r] = \frac{e^{-\mu}\mu^r}{r!}$, where $\mu = np$ (Poisson distribution)

$$a = \frac{x - \mu}{\sigma}$$ (Normal distribution)

Statistics:

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$ (Dataset mean)

$$s = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}} \quad \text{or} \quad s^2 = \frac{\sum_{i=1}^{n} x_i^{\,2} - n\left(\bar{x}\right)^2}{n-1}$$ (Dataset standard deviation)

$$\bar{x} = \frac{\sum_{i=1}^{n} f_i m_i}{\sum_{i=1}^{n} f_i}$$ (Grouped data mean)

$$s = \sqrt{\frac{\sum_{i=1}^{n} f_i (m_i - \bar{x})^2}{\sum_{i=1}^{n} f_i - 1}} \quad \text{or} \quad s^2 = \frac{\sum_{i=1}^{M} f_i m_i^{\,2} - M\left(\bar{x}\right)^2}{M-1}$$ (Grouped standard deviation)

Derivation:
First Principles: $\frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$ **Power Rule:** $\frac{dy}{dx} = nx^{n-1}$

Product Rule:

$$\frac{df}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

Quotient Rule:

$$\frac{df}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

Chain Rule:

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$$

Integral: $\int ax^n dx = \frac{ax^{n+1}}{n+1} + c, \; n \neq -1$

Definite Integral:

$$\int_a^b f'(x)dx = f(x)\Big]_a^b = f(b) - f(a)$$

Table of Probabilities – P[$Z > a$], where Z is the standard normal variable.

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
3.5	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002
3.6	0.0002	0.0002	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.7	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.8	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
4.0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000